

Pocket Money Assistant

Behavior-based spending tracker for university students

Student Name: Li Siyuan

Student ID: 24631914

Final presaentation



The Problem

💸 Out of Money

70% of students run out of funds before month-end

iiMedia Research. (2024, August 14).

📊 No Actionable Insight

Recording alone doesn't change spending habits

❓ No Classification

Can't distinguish **necessary** vs **unnecessary** expenses

Students need a tool that makes them think before they spend – not just track after

Goals & Tech Stack

4 Core Goals

- **Forced classification** of every expense
- **Budget warnings** — yellow at 80%, red at 100%
- **Visual reports** — pie chart + text summary
- **Spending awareness** through data-driven feedback

Tech Stack

Java + Swing

Desktop GUI framework

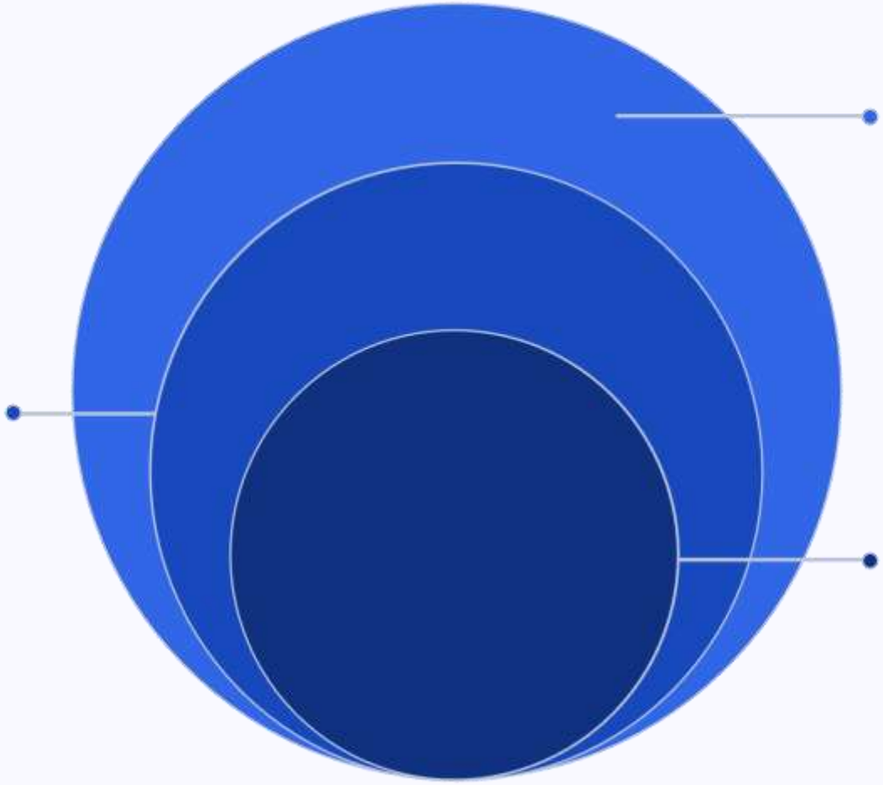
JFreeChart

Pie chart rendering

Flat Files

expenses.txt · budgets.txt · settings.txt

System Architecture



Presentation

Swing GUI: user interaction and views

Business Logic

5 Manager classes handling operations

Data Storage

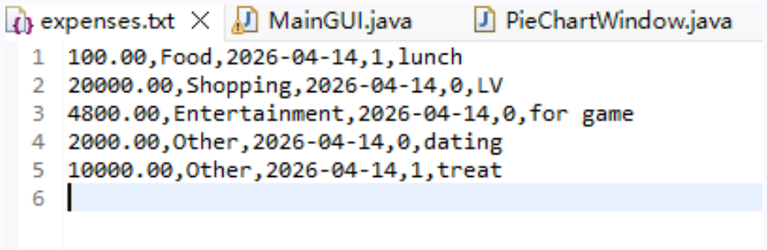
Flat files: expenses.txt budgets.txt settings.txt

A clean separation of concerns — the GUI layer delegates all logic to dedicated Manager classes, which persist data directly to flat text files.

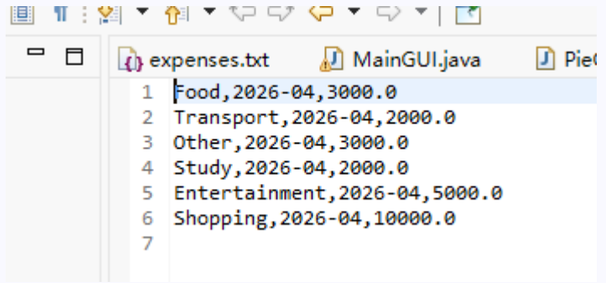
Backend Modules — 5 Managers

Manager	FR(s)	Responsibility
BudgetManager	FR5, FR6	Set monthly budgets; check status — Ok/ Warning / Exceeded
ReportManager	FR7	Generate monthly summary text report and pie chart; track view counter
FilterManager	FR8	Filter expense records by month for targeted review
EditManager	FR9	Update existing expense records in-place
DeleteManager	FR10	Delete records

Data Storage – Plain Text Files



expenses.txt (transactions)



budgets.txt (budgets)

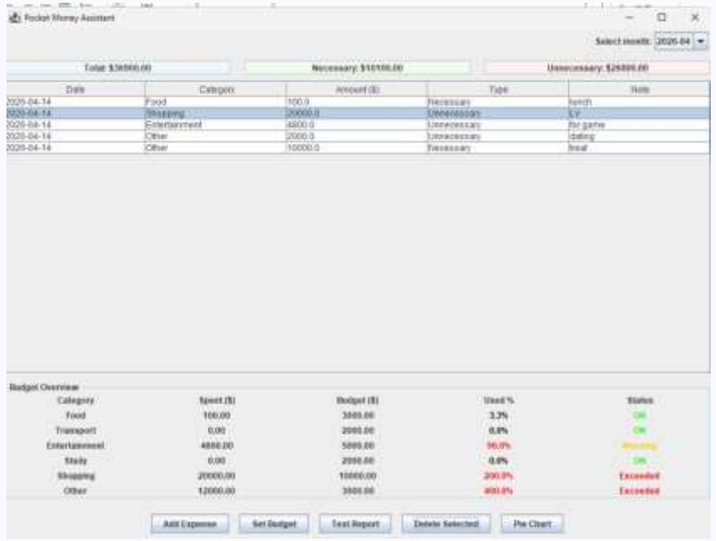
No external database required - zero configuration, easy to debug

GUI Design — Figma to Swing

Design Principles

Figma prototype → refined into Java Swing with consistent color, spacing, and component hierarchy.





🎯 Clarity

Every element has a clear purpose

✂️ Simplicity

Minimal cognitive load for students

🔁 Consistency

Uniform layout across all panels

🛡 Error Prevention

Validation before any data write

Key Features

🪟 Main Window

Dashboard overview

➕ Add Expense

Necessary + Unnecessary

⚠️ Budget Warnings

Yellow 80% · Red 100%

📊 Pie Chart

Visual spending report

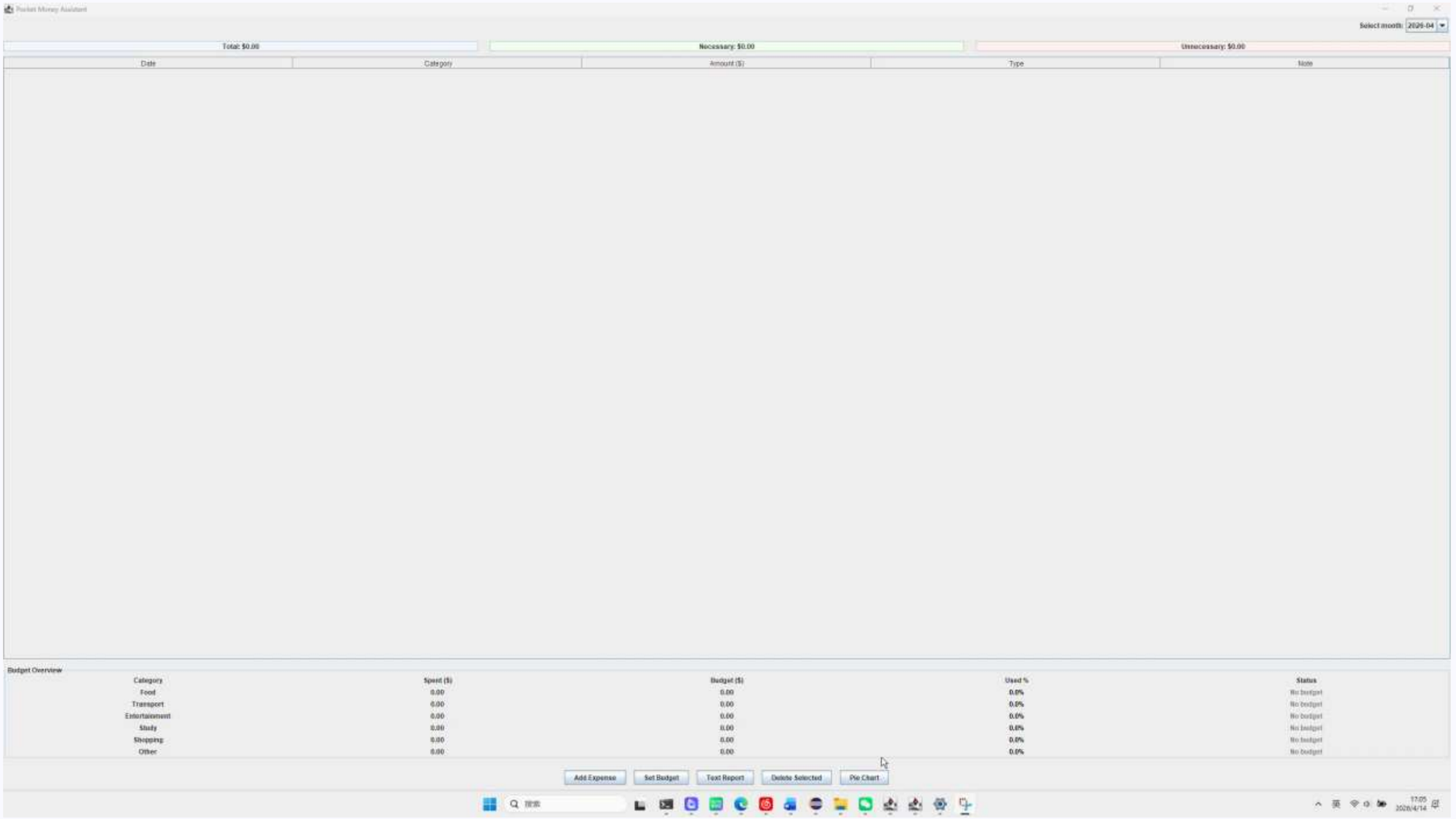
🗑 Delete

Record management

🗓 Switch Month

Historical navigation

Demonsatration



Challenges & Solutions

Swing Layout Complexity

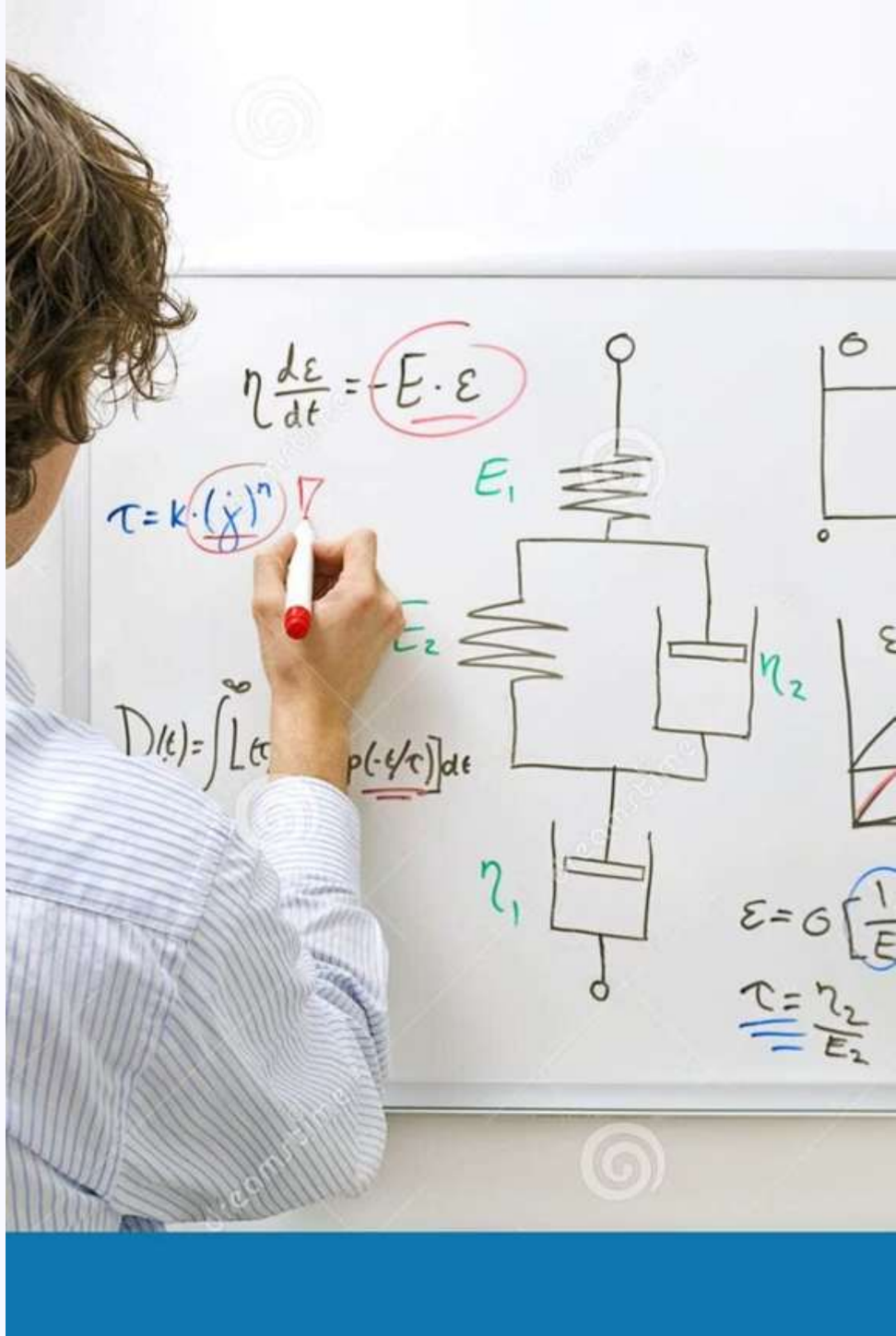
Resolved with **nested JPanels** and explicit layout managers

JFreeChart Integration

Added JAR manually to **build path** – no Maven required

Holiday Disruption

Reprioritized core FRs and leveraged buffer week to stay on schedule



User Feedback

> "Clean UI, budget warnings intuitive, pie chart extremely helpful."

> "Runs smoothly, data persistence works reliably, warnings feel natural."

Suggested Enhancements

Date Range Filter

Custom period selection

Excel Export

Portable spending reports

Progress Bars

Visual budget consumption



Conclusion & Next Steps

All 10 FRs

Fully implemented and tested

Date Filter

Custom range queries

Export

Excel / PDF output

📊 Progress Bar

Live budget visualization

Thank you